LPs in Different Form

Converting to Standard Form

$$\begin{aligned} \min:\ & c^T x \\ \text{st:}\ & A_1 x \le b_1 \\ & \cancel{A_2 x \ge b_2} \\ & A_3 x = b_3 \end{aligned}$$

Standard Form:

$$\begin{aligned} \min:\ & c^T x \\ \text{st:}\ & Ax = b \\ & x \ge 0 \end{aligned}$$

Add slack variables: s

$$\begin{aligned} \min:\ & c^T x + 0^T s \\ \text{st:}\ & A_1 x + I s = b_1 \\ & A_3 x + 0 s = b_3 \\ & s \ge 0 \end{aligned}$$

$x_i = x_i^+ - x_i^-$, $\quad x_i^+, x_i^- \ge 0$

$$\begin{aligned} \min:\ & c^T x^+ - c^T x^- + 0^T s \\ \text{st:}\ & A_1 x^+ - A_1 x^- + I s = b_1 \\ & A_3 x^+ - A_3 x^- = b_3 \\ & x^+, x^-, s \ge 0 \end{aligned}$$

Standard Form

$$C_{sf} = \begin{bmatrix} c \\ -c \\ 0 \end{bmatrix}$$

$$A_{sf} = \begin{bmatrix} A_1 & -A_1 & I \\ A_3 & -A_3 & 0 \end{bmatrix}$$

$$b_{sf} = \begin{bmatrix} b_1 \\ b_3 \end{bmatrix}$$

Absolute values 1/2

$$\min: |x_1| + |x_2|$$

$$\text{st:} \quad a_{11}x_1 + a_{12}x_2 \le b_1$$

$$a_{21}x_1 + a_{22}x_2 \le b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 \le b_m$$

not linear

$$\min: z_1 + z_2$$

$$\text{st:} \quad z_1 \ge |x_1|$$

$$z_2 \ge |x_2|$$

$$\vdots$$

$\longrightarrow$

$$\min: z_1 + z_2$$

$$z_1 \ge x_1, \quad z_1 \ge -x_1$$

$$z_2 \ge x_2, \quad z_2 \ge -x_2$$

$$\vdots$$

linear

Absolute values 2/2

Key: min: $|x|$ . . .

This is convex.

$|x| \le a_1 z_1 + a_2 z_2$

$x \le a_1 z_1 + a_2 z_2$

$-x \le a_1 z_1 + a_2 z_2$.

$|x_1| \le 1$

$x_1 \le 1$ and $x_1 \ge -1$

max: $|x_1| + |x_2|$

st: $Ax \le b$

$\xrightarrow{\text{Try}}$

max: $z_1 + z_2$

st: $Ax \le b$

$z_1 \le |x_1|$

Example:

$|x_1| \ge 1$

$x_1 \ge 1$ or $x_1 \le -1$

$$x = x^+ - x^-, \quad x^+, x^- \geq 0$$

In some cases: $|x| = x^+ + x^-$

This is the case when either x^+ or x^- is equal to zero.

$$\begin{aligned} \min:\ & |x_1| + |x_2| \\ \text{st:}\ & Ax \leq b \end{aligned} \quad \rightarrow \quad \begin{aligned} \min:\ & (x_1^+ + x_1^-) + (x_2^+ + x_2^-) \\ \text{st:}\ & Ax^+ - Ax^- \leq b \\ & x^+, x^- \geq 0 \end{aligned}$$

Tempting but not correct to make this substitution in:

$$\begin{aligned} \max:\ & |x_1| + |x_2| \\ \text{st:}\ & Ax \leq b \end{aligned}$$

// Think about why this substitution is not correct here.